

GENETICS 101 · WEEK 2 QUIZ

Genetics 101 — Week 2

Name: _____

Date: _____

How to play: Read each question twice, read every choice, cross out the wrong ones, then put an X in the box next to your answer. Get **6 of 8** right to earn the badge.

1. When is the newborn heel-prick screen usually done?

- ☐ A. In the first minute of life, right after the Apgar score.
- ☐ B. At 24 to 48 hours of age, after the baby has started feeding.
- ☐ C. At the first checkup, about two weeks after going home.
- ☐ D. On the baby's first birthday, when the blood is easier to draw.

2. Why is the poke made on the SIDE of the heel?

- ☐ A. The side of the heel has no nerves, so the baby feels nothing.
- ☐ B. The side has plenty of blood and no bone right underneath it.
- ☐ C. The middle of the heel is too thick for the drops to come out.
- ☐ D. The side is where the umbilical cord blood collects after birth.

3. What is PKU?

- ☐ A. A wrong letter in the hemoglobin recipe bends red blood cells; early medicine keeps them from jamming.
- ☐ B. A missing hormone from the thyroid gland slows growth; a daily pill from the first weeks replaces it.
- ☐ C. A slow liver enzyme lets bilirubin pile up and turn the skin yellow; blue light clears it.
- ☐ D. A broken enzyme lets phenylalanine pile up and harm the brain; a special diet from the first weeks prevents it.

4. What is an enzyme?

- ☐ A. One of the four chemical letters that the DNA instruction book is written in.
- ☐ B. A protein whose job is a chemical task — snipping, gluing, or taking a molecule apart.
- ☐ C. A hormone that travels through the blood to give the body's organs their orders.
- ☐ D. A kind of sugar that the mitochondria burn with oxygen to make energy for the cell.

5. How did Robert Guthrie's 1961 test spot PKU from a drop of blood?

- ☐ A. The blood spot was fed to mice, and the mice with PKU blood stopped growing.
- ☐ B. A chemical dye turned the blood spot bright yellow when phenylalanine was present.
- ☐ C. Bacteria that need phenylalanine grew in a fat ring around blood spots with too much of it.
- ☐ D. A microscope showed the red blood cells bent into sickle shapes in a PKU baby.

6. PKU happens in only about 1 baby in 12,000. Why does every baby get tested anyway?

- ☐ A. It hides at birth, waiting causes permanent harm, early treatment works, and the test is cheap.
- ☐ B. Doctors can't tell which babies are at risk, so testing everyone is a legal requirement.
- ☐ C. The test also measures the baby's blood type, which every hospital needs to know.
- ☐ D. The heel prick also gives the baby a small dose of the medicine that prevents PKU.

7. Which is the MOST common condition the heel-prick screen catches?

- ☐ A. Galactosemia – an enzyme that can't handle milk sugar, fixed by milk-free feeding.
- ☐ B. PKU – a broken enzyme, fixed with a special low-phenylalanine diet for life.
- ☐ C. Cystic fibrosis – thick mucus in the lungs and gut, treated every single day.
- ☐ D. Congenital hypothyroidism – too little thyroid hormone, fixed with a daily pill.

8. Which two newborn screens do NOT use blood?

- ☐ A. The hearing test and the pulse-oximeter heart test.
- ☐ B. The PKU test and the thyroid test.
- ☐ C. The Apgar score and the weight on the growth chart.
- ☐ D. The brain ultrasound and the bilirubin light.

Score: _____ / 8

Badge earned? YES / NOT YET (need 6)